

# CSCI-246 Discrete Structures HW 05

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## Objective

- Understanding mathematical induction.
- Understanding graph definitions.
- Understanding the problem solving process.
- Understanding summation notation.

## Submission requirements

- *Type or clearly hand-write* your solutions into a **PDF FORMAT**.
- **DO NOT UPLOAD images.**
- *non-pdf or emailed solutions will not be graded.*
- If you take pictures of your handwritten homework, put it into pdf format.
- *Start each problem in a new page.*
- Follow the model that you have learned during the lectures for proofs.
- Do not wait until the last minute to submit the assignment.
- You can submit any number of times before the deadline.

- If you are using latex, and you do not know how to type a symbol, use the following website. You can draw the symbol here and it will give you the latex code and the packages that you have to import. <https://detexify.kirelabs.org/classify.html>
- If you are using latex to write the answer, you can use overleaf to make your life easier. **Overleaf is a free, online platform that helps users create and publish scientific and technical documents using LaTeX, a markup-based document preparation system**
- If you do not understand a problem, ask questions during/after the lectures, or during office hours or via discord.
- ***Do not use generative AI to write answers.***

## 1 Q1

Write the following sequences using summation notation. (This is an easy question, don't try to complicate this. I am not asking to give an expression, I am asking you to write the sequences using summation notation.)

- $0 + 3 + 6 + 9 + \cdots + 3 \cdot i + \cdots + 3 \cdot n$ . (Sum of the first  $n + 1$  natural numbers.)
- $0^3 + 1^3 + 2^3 + \cdots + i^3 + \cdots + n^3$ . (Sum of the first  $n + 1$  cubes of natural numbers).
- $2^2 + 4^2 + 6^2 + \cdots + (2i)^2 + \cdots + (2 \cdot n)^2$ . (Sum of the squares of first  $n$  even numbers starting at 2.)

## 2 Q2

Draw a graph with the nodes  $V = \{1, 2, \dots, 9, 10\}$ , and edges between  $x \in V$  and  $y \in V$  if  $x$  divides  $y$ . Does it make sense to use a directed or undirected graph? Is the graph that you have drawn simple?

## 3 Q3

In the class we looked at a property of an undirected graph  $G = (V, E)$ .

$$\sum_{v \in V} \deg(v) = 2 \cdot |E|$$

This property is known as **Handshaking Lemma**.

Here we will try to prove this using induction. The task for you is to fill in the missing pieces of this proof.

**Proof:** Here, I define the graph  $G$  as a recursively defined structure.

Undirected Graph  $G$  can be defined as follows:

- Base case: Start with an graph with no edges and **any number of isolated vertices** (each with degree zero).
- Recursive case: Add an edge between two vertices in the graph, updating the degrees of those two vertices.

Now that we have defined the undirected graph as a recursively defined structure, we can move to induction steps as follows:

For any natural number  $m \geq 0$ , let predicate  $P(m)$  be true if any graph  $G = (V, E)$  with  $m$  edges has the following property:  $\sum_{v \in V} \deg(v) = 2 \cdot m$ — false otherwise. We show that  $\forall m \geq 0 : P(m)$  using mathematical induction over  $m$ .

Base case:

**You have to fill the base case and the proof of base case here.**

Inductive case:

We want to show that  $\forall m \geq 1 : P(m - 1) \implies P(m)$ . For the inductive hypothesis we assume that  $P(m - 1)$  is true; that is, we assume that **any** undirected graph  $G = (V, E)$  with  $m - 1$  edges has  $\sum_{v \in V} \deg(v) = 2 \cdot (m - 1)$ . Now let  $H = (W, F)$  be any graph with  $m$  edges. We want to prove that  $P(m)$  holds; that is,  $\sum_{v \in W} \deg(v) = 2 \cdot m$ .

Let  $\{u, w\}$  be any edge of  $H$ . Let us consider the graph  $G$  constructed from  $H$  by removing that one edge  $\{u, w\}$  from  $H$ ; that is,  $G = (W, F \setminus \{\{u, w\}\})$  (basically  $G$  is the graph that is obtained by removing the edge  $\{u, w\}$ ). Note that we keep the same set of vertices; we only remove a single edge. For any vertex  $v \in W$ , let  $\deg_G(v)$  denote the degree of  $v$  in graph  $G$ , and  $\deg_H(v)$  denote the degree in graph  $H$ , since these could be different once you remove the edge  $\{u, w\}$ .

**Fill the rest of the proof of the inductive case here.**

Since we showed that  $P(0)$ , and  $\forall m \geq 1 : P(m - 1) \implies P(m)$ , we can conclude that  $\forall m \geq 0 : P(m)$  using the principle of mathematical induction.

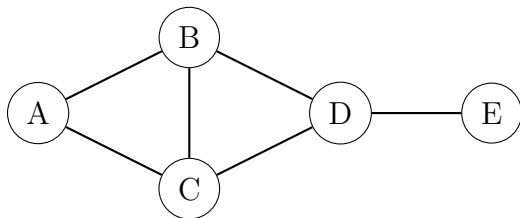
## 4 Q4

In the lecture on special graphs, we proved that if a graph contains a triangle, then it is not bipartite.

1. What is the converse of this statement?
2. What is the inverse of this statement?
3. Is the converse of the statement true or false?
4. Prove your answer to the previous question.

## 5 Q5

In the lectures, we learned that distance between two nodes is the length of the shortest path between them. The **Diameter** of a graph is the largest distance between any two nodes in the graph. For example, for the following graph, the diameter is 3, which is the distance between  $A$  and  $E$ . One of the shortest paths between  $A$  and  $E$  is  $(A, B, D, E)$ .



1. What could be the largest diameter of a connected graph with  $n$  nodes can have and **explain your answer**.
2. What could be the smallest diameter that a connected graph with  $n$  nodes can have and explain the answer?
3. Draw the above two graphs for  $n = 5$ .

## 6 Q6

1. Suppose that Alice has exactly 160 friends, and each of her friends has exactly 160 friends—that is, a friend of Alice knows Alice and 159 **other** people. (Note that Alice's friends' sets

of friends can overlap.) Let  $S$  denote the set of people that Alice knows directly or with whom Alice has a mutual friend. What's the largest possible value of  $|S|$ ? **Explain how you got this answer using graph based approach.**

**Hint: Think of these friendships in terms of a graph. You can create a node for each person and connect them with an undirected edge if they are friends. How can you create a graph such that the set  $S$  is maximized—If you answer this question then you can come up with an answer to this question?**

2. For the set  $S$  defined above, what's the smallest possible value of  $|S|$ ? Explain what type of graph would that be if people are defined as nodes and friendships are defined as an undirected edge.
3. Let  $u$  be a node in an undirected graph  $G$ . Prove that  $u$ 's degree is at most the sum of the degrees of  $u$ 's neighbors. Hint: use direct proof.